

AI508

Computer

Vision

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Tasks of the day

Image classification

Object detection

Segmentation

Image generation

Inpainting

Image-text retrieval

Visual question answering

Video generation

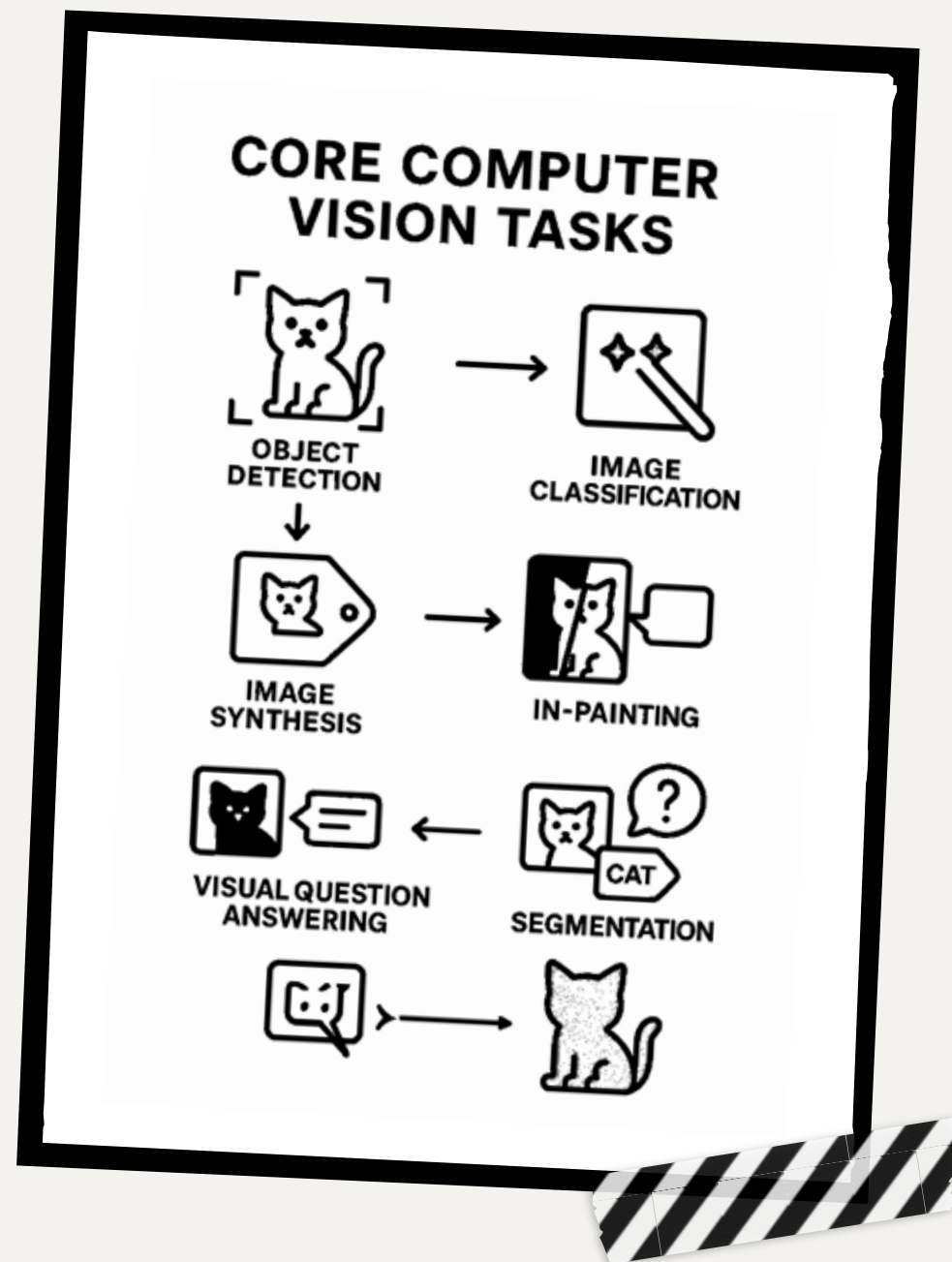


Image generation

GOAL:

Synthesize realistic images.

METRICS:

FID, KID, CLIPScore; human preference.

DATASETS:

LAION-5B, COCO, ImageNet.

ARCHITECTURES:

Diffusion, VAE, GAN.



Inpainting

GOAL:

Fill masked regions coherently.

METRICS:

PSNR, SSIM, LPIPS.

DATASETS:

Places2, CelebA-HQ.

ARCHITECTURES:

Diffusion, Partial/Gated/Dilated CNNs.



Generative models

- From observed samples x learn data distribution $p(x)$
- Generate new samples from $p(x)$
- Often, generated from latent variable z
- With joint distribution $p(x,z)$, we have two ways of obtaining $p(x)$:
 1. $p(x) = \int p(x,z) dz$
 2. $p(x) = p(x,z) / p(z|x)$
- Integrating z out (Option 1) is intractable for complex models
- Using the chain rule of probability (Option 2) requires access to a ground truth encoder $p(z|x)$.



Evidence Lower Bound (ELBO)

- We need to approximate the ground truth latent variable encoder $p(z|x)$
- We denote a variational distribution with parameters ϕ by $q_\phi(z|x)$
- Then the ELBO is a lower bound of the log likelihood of the observed data:

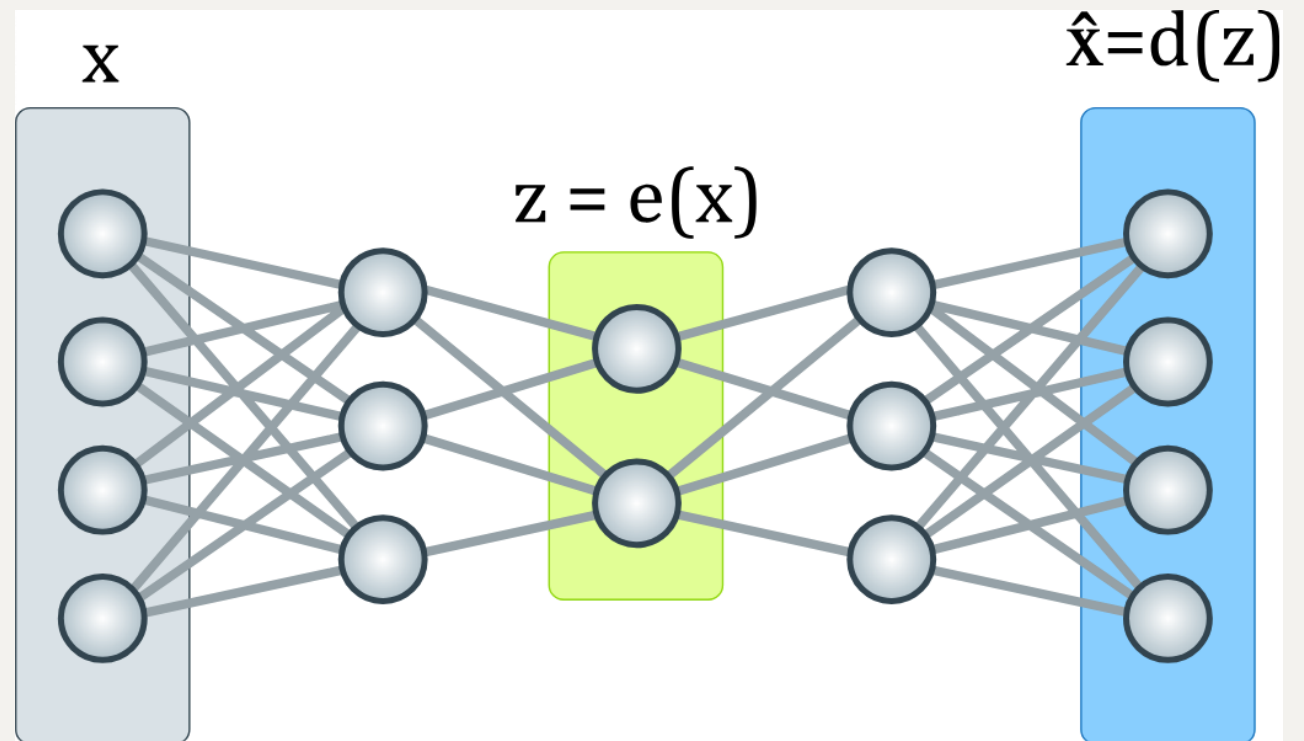
$$\mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p(x, z)}{q_\phi(z|x)} \right]$$

- As $\log p(x) \geq \text{ELBO}$, maximizing ELBO approximates the encoder



Autoencoder

- Encode inputs to a (typically lower-dimensional) latent representation
- Decode the latent representation to the same dimension as the input
- Trained by teaching the model to reconstruct inputs
- Can be seen as a form of lossy compressions
- Applications in denoising, medical image reconstruction, superresolution and many other computer vision tasks

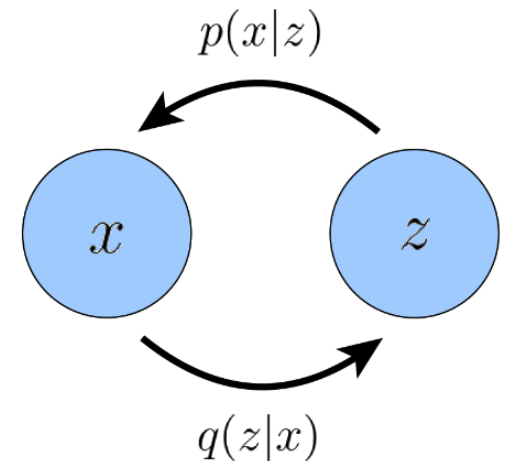


Variational autoencoders (VAEs)

- Directly maximize ELBO
- Variational as we optimize for the best $q_\phi(z|x)$ for any set of parameters ϕ
- Encoder $q_\phi(z|x)$ transforms observation x into a latent vector z
- Decoder $p_\theta(x|z)$ transforms a latent vector z into an observation x

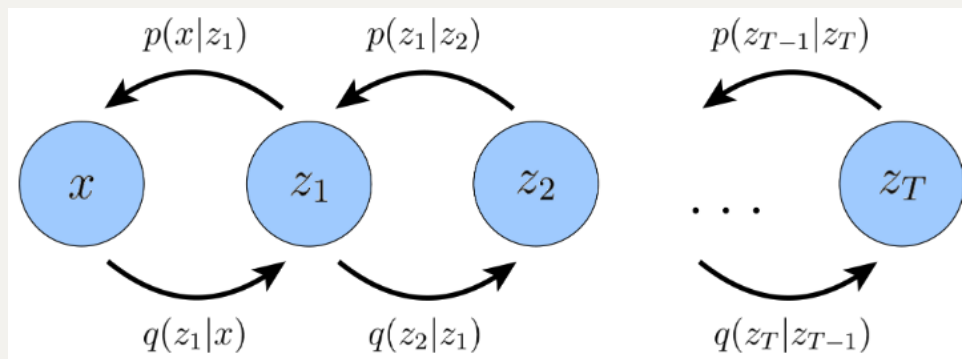
$$\begin{aligned}\mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p(x, z)}{q_\phi(z|x)} \right] &= \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p_\theta(x|z)p(z)}{q_\phi(z|x)} \right] \\ &= \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] + \mathbb{E}_{q_\phi(z|x)} \left[\log \frac{p(z)}{q_\phi(z|x)} \right] \\ &= \underbrace{\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)]}_{\text{reconstruction term}} - \underbrace{D_{\text{KL}}(q_\phi(z|x) \parallel p(z))}_{\text{prior matching term}}\end{aligned}$$

<https://arxiv.org/pdf/2208.11979>



Markovian hierarchical VAEs

- A generalization of VAEs with multiple levels of latent variables
- General hierarchical VAEs allow connection to all previous levels



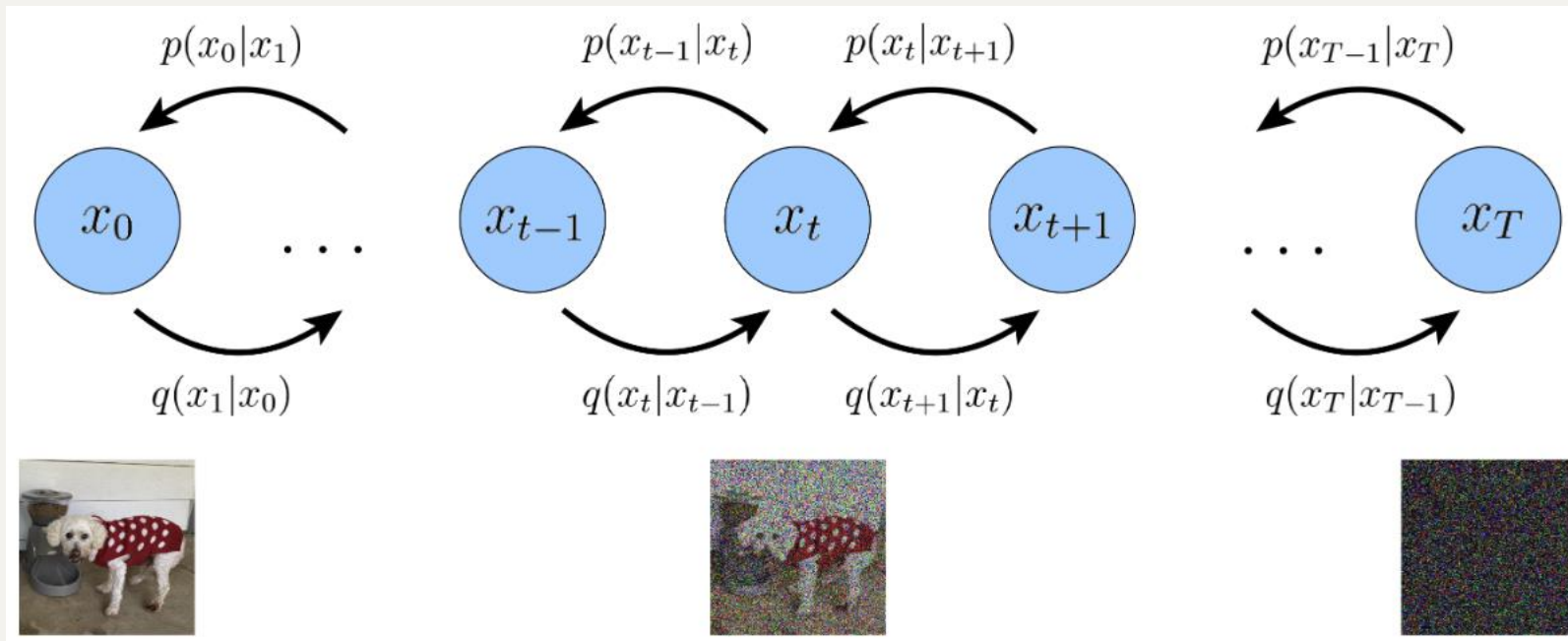
- Markovian VAEs restrict to one-step transitions for both encoders and decoders
- The joint encoder and decoder then become:

$$p(\mathbf{x}, \mathbf{z}_{1:T}) = p(z_T) p_{\theta}(\mathbf{x} | z_1) \prod_{t=2}^T p_{\theta}(z_{t-1} | z_t)$$
$$q_{\phi}(\mathbf{z}_{1:T} | \mathbf{x}) = q_{\phi}(z_1 | \mathbf{x}) \prod_{t=2}^T q_{\phi}(z_t | z_{t-1})$$



Variational diffusion models

- A Markovian hierarchical VAE with the following three characteristics:
 1. Latent dimensions == data dimension
 2. The encoder is parameter-free and each step simply adds Gaussian noise
 3. The amount of noise added varies such that x_T is standard Gaussian distributed



Conditional diffusion models

- We rarely want to generate completely random images
- Instead of $p(x)$, we are often interested in $p(x|y)$
- Here, y is some type of conditioning information
- Examples of conditioning information:
 - + Text embedding
 - + Low-resolution image
 - + Image with masked parts
- Make encoder steps $p_\phi(x_{t-1}|x_t)$ conditional:
 $p_\phi(x_{t-1}|x_t, y)$
- Such conditional diffusion models may learn to ignore conditioning information :-)



Classifier guidance

- At each step t , classify the noise latent $\mathbf{x}_t$ using a classifier
- Classifier estimates to what degree $\mathbf{x}_t$ conforms to conditioning information y
- Then, the gradient of the conditional log likelihood is:

$$\nabla \log p(\mathbf{x}_t|y) = \nabla \log p(\mathbf{x}_t) + \gamma \nabla \log p(y|\mathbf{x}_t)$$

- Classifier guidance assumes the existence of a good classifier
- The classifier needs to work with all levels of noisy images
- In practice, this requires training on T versions of images ☹



Classifier-free guidance

- Combine a conditional and an unconditional diffusion model:

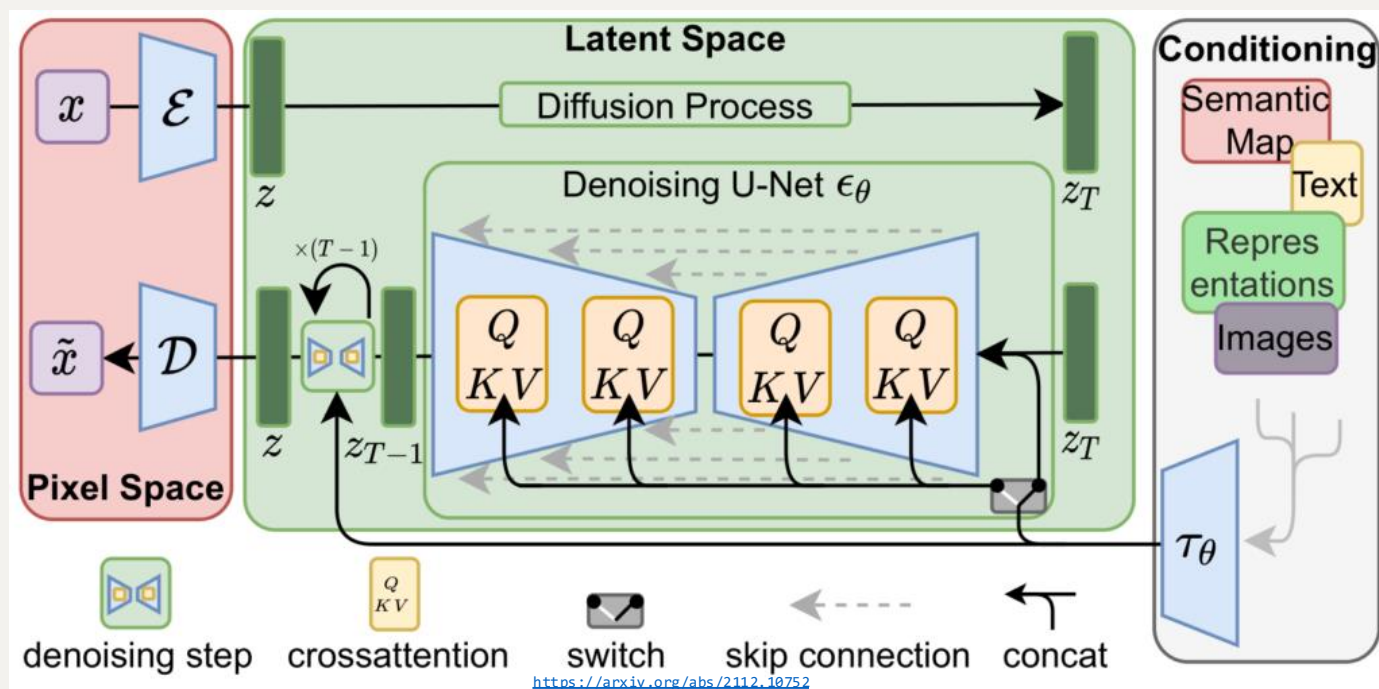
$$\begin{aligned}\nabla \log p(\mathbf{x}_t|y) &= \nabla \log p(\mathbf{x}_t) + \gamma (\nabla \log p(\mathbf{x}_t|y) - \nabla \log p(\mathbf{x}_t)) \\ &= \nabla \log p(\mathbf{x}_t) + \gamma \nabla \log p(\mathbf{x}_t|y) - \gamma \nabla \log p(\mathbf{x}_t) \\ &= \underbrace{\gamma \nabla \log p(\mathbf{x}_t|y)}_{\text{conditional score}} + \underbrace{(1 - \gamma) \nabla \log p(\mathbf{x}_t)}_{\text{unconditional score}}\end{aligned}$$

- Here, λ controls the importance attached to the conditioning (strength)
- In practice, we learn one (conditional) model instead of two
- The conditional model is taught to ignore all-zero conditioning input
- This corresponds to random dropout on the conditioning information



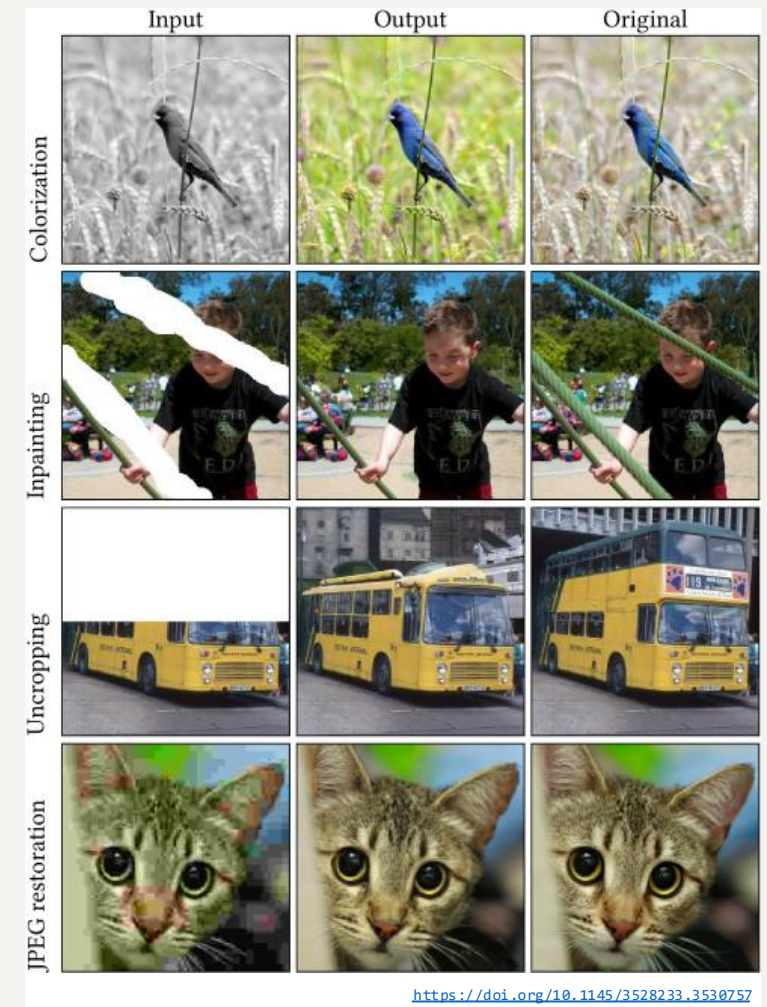
Image generation via conditional VDM

- Use text embeddings as conditioning information
- Most prominent model is probably Stable Diffusion (SD)
- Instead of directly embedding text, SD uses separate text-image embeddings (CLIP)
- Text embeddings are made available to the UNet through cross attention



Inpainting via conditional VDM

- Use image as conditioning information
- Parts of the images to be inpainted are obstructed
- At each step t of the denoising process, the conditioning information keeps the image grounded



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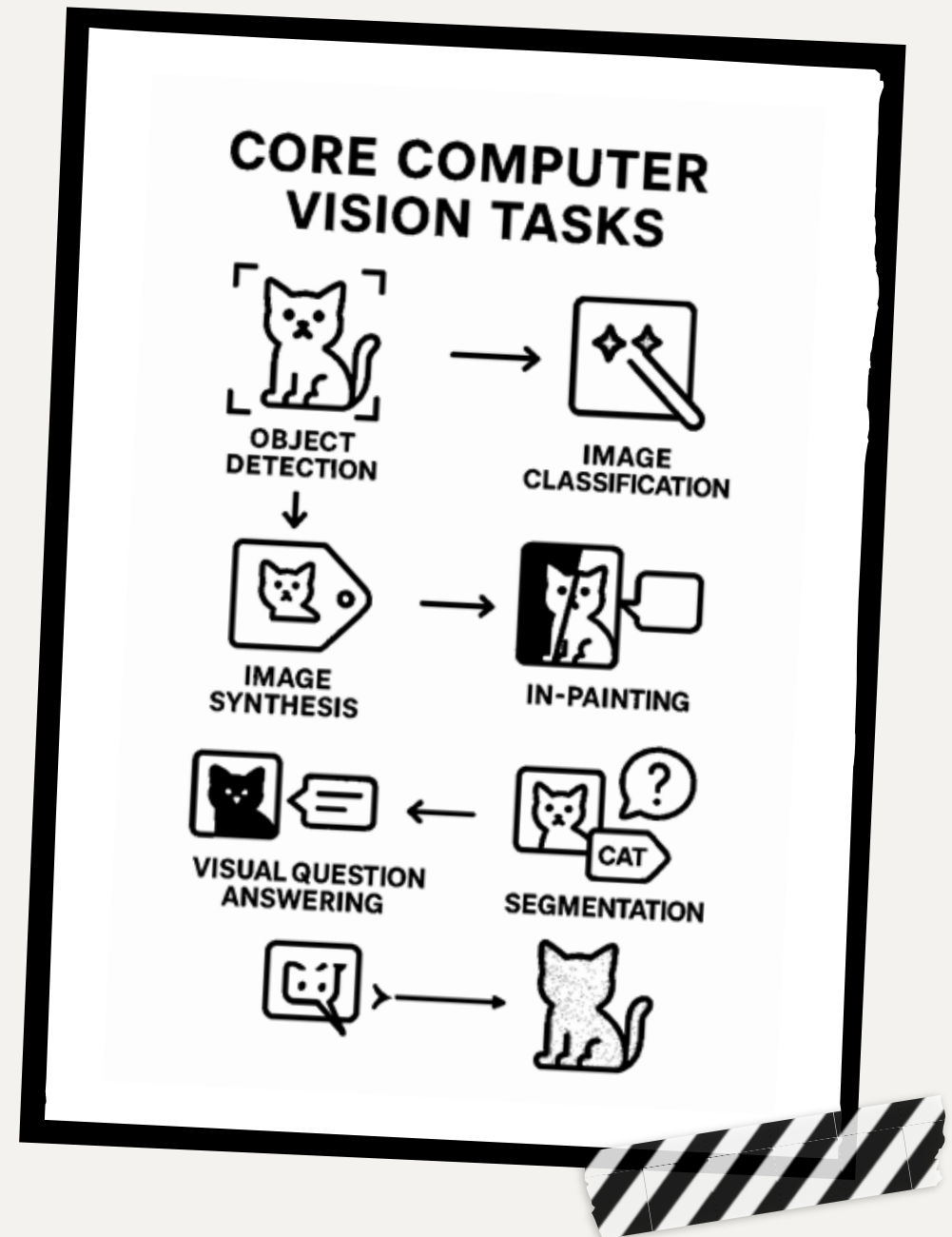


Image-text retrieval

GOAL :

Embed images and text jointly.

METRICS:

Recall, mean Average Precision.

DATASETS:

CLIP, LAION.

ARCHITECTURES:

CLIP, BLIP, ViLBERT.

	Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet		76.2	76.2	0%
ImageNetV2		64.3	70.1	+5.8%
ImageNet-R		37.7	88.9	+51.2%
ObjectNet		32.6	72.3	+39.7%
ImageNet Sketch		25.2	60.2	+35.0%
ImageNet-A		2.7	77.1	+74.4%



Visual question answering

GOAL :

Answer questions regarding an image.

METRICS:

Accuracy, LLM-as-a-judge.

DATASETS:

VQAv2, GQA, TextVQA, OK-VQA.

ARCHITECTURES:

VLM (CLIP/BLIP + LLM).



What does this image show?

You're standing in Plaza del Ayuntamiento, Valencia, in front of the Ayuntamiento (City Hall). Blue hour shot, strong mixed lighting, wide-angle phone look, heavy highlight clipping on the façade.

